MTH2010 Algebra

Problem Sheet 1: Solutions

In these solutions whenever we have a group G, or a homomorphism $f : G \to H$, we omit the explicit mention of the group law of G (and H) and denote these group laws simply as a product: if $g_1, g_2 \in G$ then $g_1 g_2$ denotes the composition of g_1 and g_2 in the given order.

1. (a) $\mathbf{Z}_8 = \{[0], [1], [2], [3], [4], [5], [6], [7]\}$.
In $\mathbf{Z}_8$ the order of elements are: $\text{ord}[0] = 1$, $\text{ord}[1] = \text{ord}[3] = \text{ord}[5] = \text{ord}[7] = 8$, $\text{ord}[2] = \text{ord}[6] = 4$, and $\text{ord}[4] = 2$ (use the formula given in lecture for the order of elements of $\mathbf{Z}_n$).
Similarly:
$\mathbf{Z}_{12} = \{[0], [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11]\}$.
In $\mathbf{Z}_{12}$ the order of elements are: $\text{ord}[0] = 1$, $\text{ord}[1] = \text{ord}[5] = \text{ord}[7] = \text{ord}[11] = 12$, $\text{ord}[2] = 6$, $\text{ord}[6] = 2$, $\text{ord}[4] = \text{ord}[8] = 3$, $\text{ord}[3] = \text{ord}[9] = 4$, and $\text{ord}[10] = 6$.

(b) Since the order of an element of D_{2n} must divide the cardinality of the group D_{2n}, which is $2n$, the possibilities for the order of such an element are all positive integer divisors of $2n$, which are: 1, 2, d and $2d$ where $d > 1$ is any divisor of n.

- An element of D_{2n} which is not a power of r has the form sr^k and we have $(sr^k)^2 = sr^k sr^k = ssr^{n-k}r^k = s^2 r^{n-k+k} = s^2 r^n = s^2 = e$ since s has order 2. Thus sr^k has order 2.
- Assume n is odd and $d > 2$ divides n. An element of D_{2n} is either a power of r in which case its order divides $n = \text{ord}(r)$ (which is the cardinality of the cyclic subgroup of D_{2n} generated by r) hence is odd, or is of the form sr^k in which case it has order 2. So D_{2n} has no elements of order $2d$, and a fortiori has no cyclic subgroup of that cardinality as well.

3. (a) (i) The cyclic subgroups of $D_8 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$ are the subgroups generated by elements of D_8 which are:
$\{1\}$ subgroup of cardinality 1,
$\langle r \rangle = \langle r^3 \rangle = \{1, r, r^2, r^3\}$ subgroup of cardinality 4,
$\langle s \rangle = \{1, s\}$, $\langle r^2 \rangle = \{1, r^2\}$, $\langle sr \rangle = \{1, sr\}$, $\langle sr^3 \rangle = \{1, sr^3\}$, and $\langle sr^2 \rangle = \{1, sr^2\}$ subgroups of cardinality 2.

- The subgroup $\{e, r^2, s, sr^2\}$ is a subgroup of D_8 (check it is a subgroup) of cardinality 4 which is not cyclic as it contains no element of order 4 [a necessary condition for a group of cardinality 4 to be cyclic] as $\text{ord}(r^2) = \text{ord}(s) = \text{ord}(sr^2) = 2$.

(ii)
- The subset $\{e, r^4, s, sr^4\} \subset D_{16}$ is a subgroup (check it is a subgroup) of cardinality 4. It is not a cyclic subgroup as it contains no element of order 4: a necessary condition for a group of cardinality 4 to be cyclic. Indeed, $\text{ord}(r^4) = \text{ord}(s) = \text{ord}(sr^4) = 2$.
- The subset $\{e, r^2, r^4, r^6, s, sr^2, sr^4, sr^6\} \subset D_{16}$ is a subgroup (check it is a subgroup) of cardinality 8. It is not cyclic as it contains no element of order 8: a necessary condition for a group of cardinality 8 to be cyclic. Indeed $\text{ord}(r^4) = \text{ord}(sr^2) = \text{ord}(sr^4) = \text{ord}(sr^6) = 2$, and $\text{ord}(r^2) = \text{ord}(r^6) = 4$.

(b) Let $m = \frac{n}{d}$. Then $\langle r^m \rangle = \{e, r^m, r^{2m}, \ldots, r^{(d-1)m}\}$ is a cyclic subgroup of D_{2n} of cardinality d since $\text{ord}(r^m) = d$ (check this fact).

- The subgroup $\{e, r^m, r^{2m}, \dots, r^{(d-1)m}, s, sr^m, sr^{2m}, \dots, sr^{(d-1)m}\}$ (check this is a subgroup) is a subgroup of D_{2n} of cardinality $2d$. It is not cyclic as it contains no element of order $2d$: the order of r^{km} divides the cardinality of $\langle r^m \rangle$ which is d, and the order of an element sr^{km} is 2.

4. (a) Let H be a subgroup of $G = \langle x \rangle$ and $n \geq 1$ the smallest positive integer such that $y = x^n \in H$ (convince yourself such n exists). We show $\langle y \rangle = H$ is cyclic generated by y. Clearly $\langle y \rangle \subset H$ since $y \in H$ and H is a subgroup. We prove $H \subset \langle y \rangle$. Let $z \in H$, $z = x^m$ for some $m \geq 1$. Then $m \geq n$ by the definition of n. We have $m = kn + r$ with $0 \leq r < n$ (Euclidian division). Hence $x^m = x^{kn+r} = x^{kn}x^r = (x^n)^k x^r = y^k x^r$ which implies $x^r = y^{-k}x^m \in H$ since $x^m, y^{-k} \in H$, and H is a subgroup. But then $r = 0$ by the definition of n and $z = x^m = y^k \in \langle y \rangle$. If $z = x^m \in H$ with $m < 0$, then $z^{-1} = x^{-m} \in \langle y \rangle$ by the above discussion (note that $z^{-1} \in H$ since H is a subgroup), and $z \in \langle y \rangle$ since $\langle y \rangle$ is a subgroup.

(b) Assume $G = \langle x \rangle$ is cyclic with $\text{ord}(x) = n$ and $d \geq 1$ is a divisor of n. Then $H = \langle x^{\frac{n}{d}} \rangle$ is a subgroup of G of cardinality d [check that $\text{ord}(x^{\frac{n}{d}}) = d$, alternatively see Properties/Examples 1.3, item 5, in the Week 1 lecture notes].
Let H' be a subgroup of G of cardinality d, we show $H' = H = \langle x^{\frac{n}{d}} \rangle$. It holds $H' = \langle x^m \rangle$, where $1 \leq m \leq n-1$ is the smallest integer $1 \leq t \leq n-1$ such that $x^t \in H'$ [see part (a)]. Further $|H'| = \text{ord}(x^m) = d = \frac{n}{\gcd(n,m)}$ [see Properties/Examples 1.3, item 5, in the Week 1 lecture notes]. Hence $\gcd(n, m) = \frac{n}{d}$, and $\frac{n}{d}k = m$ for some $1 \leq k \leq m$. Thus $x^m = x^{\frac{n}{d}k} = (x^{\frac{n}{d}})^k \in H$ and $H' \subset H$ [if $y = x^{tm} \in H'$ then $y = (x^{\frac{n}{d}})^{tk} \in H$]. Since $|H| = |H'| = d$, it must hold that $H = H'$.

5. (a) If A and B are abelian then clearly $A \times B$ is abelian as follows easily from the definition of the composition law in $A \times B$: $(a_1, b_1)(a_2, b_2) = (a_1a_2, b_1b_2) = (a_2a_1, b_2b_1) = (a_2, b_2)(a_1, b_1)$. Now assume that $A \times B$ is abelian we show A is abelian. Let $a_1, a_2 \in A$. Then $(a_1, e_B), (a_2, e_B) \in A \times B$ and $(a_1, e_B)(a_2, e_B) = (a_1a_2, e_B) = (a_2, e_B)(a_1, e_B) = (a_2a_1, e_B)$ which implies $a_1a_2 = a_2a_1$ hence A is abelian. The proof that B is abelian is entirely similar.

- Suppose $A \times B = \langle (a, b) \rangle$ is cyclic generated by (a, b), we show that $A = \langle a \rangle$ and $B = \langle b \rangle$. If $\tilde{a} \in A$ then $(\tilde{a}, e_B) \in A \times B$, hence $(\tilde{a}, e_B) = (a, b)^k = (a^k, b^k)$ for some $k \geq 1$, which implies $\tilde{a} = a^k$ is a power of a. This shows that A is cyclic generated by a. The proof that $B = \langle b \rangle$ is cyclic generated by b is entirely similar.
The converse is not true: $\mathbb{Z} = \langle 1 \rangle$ is cyclic, but $\mathbb{Z} \times \mathbb{Z}$ is not cyclic. Indeed, assume $\mathbb{Z} \times \mathbb{Z} = \langle (a, b) \rangle$ is cyclic generated by the element (a, b). Then every $(m, n) \in \mathbb{Z} \times \mathbb{Z}$ equals (ka, kb) for some $k \in \mathbb{Z}$, in particular both a and b are non zero. But this is not possible: take m to be an integer which is not a multiple of a, or n an integer which is not a multiple of b, then $(m, n) \neq (ka, kb)$.

(b) $e_A \in C$ and $e_B \in D$ as these are subgroups hence $(e_A, e_B) \in C \times D$. The fact that $C \times D$ is closed under the composition law of $A \times B$ follows easily from the definition of this law and the fact that C and D are subgroups: if $c_1, c_2 \in C$, $d_1, d_2 \in D$ then $(c_1, d_1)(c_2, d_2) = (c_1c_2, d_1d_2) \in C \times D$ since $c_1c_2 \in C$ and $d_1d_2 \in D$. Also if $(c, d) \in C \times D$ then $(c, d)^{-1} = (c^{-1}, d^{-1}) \in C \times D$ since $c^{-1} \in C$ and $d^{-1} \in D$, C and D being subgroups of A and B respectively.

- The subset $H = \{(a, a) \ : a \in \mathbb{Z}\}$ is a subgroup of $\mathbb{Z} \times \mathbb{Z}$, it is the kernel of the homomorphism $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ defined by $(a, b) \mapsto a - b$ (check this is a homomorphism). But $H \neq m\mathbb{Z} \times n\mathbb{Z}$ for any possible integers n and m, as for $t, s \in \mathbb{Z}$ such that $tn \neq$

$sm, (ms, tn) \in m\mathbb{Z} \times n\mathbb{Z}$ but $(ms, tn) \notin H$.

(c) $(a, b) = (a, e_B) \times (e_A, b) = (e_A, b) \times (a, e_B)$ hence $(a, b)^k = [(a, e_B) \times (e_A, b)]^k = (a, e_B)^k \times (e_A, b)^k = (a^k, e_B) \times (e_A, b^k), \forall k \geq 1$ since (a, e_B) and (e_A, b) commute (prove this fact). From this we deduce that $\text{ord}(a, b) = \text{lcm}(\text{ord}(a, e_B), \text{ord}(e_A, b)) = \text{lcm}(\text{ord}(a), \text{ord}(b))$ [note that $\text{ord}(a, e_B) = \text{ord}(a) = n$, and $\text{ord}(e_A, b) = \text{ord}(b) = m$]. Indeed let $(a, b)^\alpha = (a^\alpha, b^\alpha) = (e_A, e_B)$ then $a^\alpha = e_A$; which implies $n = \text{ord}(a)$ divides α, and $b^\alpha = e_B$; which implies $m = \text{ord}(b)$ divides α, hence $\text{lcm}(\text{ord}(a), \text{ord}(b))$ divides α. On the other hand $(a, b)^{\text{lcm}(n,m)} = (a^{\text{lcm}(n,m)}, b^{\text{lcm}(n,m)}) = (e_A, e_B)$, so that $\text{ord}((a, b)) = \text{lcm}(n, m)$.

- Next suppose $\text{ord}(a) = \infty$ then $\text{ord}(a, b) = \infty$. For otherwise suppose $\text{ord}(a, b) = k$ is finite, then $(a, b)^k = (a^k, b^k) = (e_A, e_B)$, hence $a^k = e_A$ which contradicts $\text{ord}(a) = \infty$.

6. (a) First consider the case $n \geq 0$. Let $x \in G$. We argue by induction. The case $n = 0$ holds as $\phi(x^0) = \phi(e_G) = e_H = \phi(x)^0$.
Now let $k \geq 0$ be an integer. Suppose $\phi(x^k) = \phi(x)^k$. Then $\phi(x^{k+1}) = \phi(x^k.x) = \phi(x^k)\phi(x)$, as ϕ is a homomorphism. Continuing, $\phi(x^{k+1}) = \phi(x)^k\phi(x) = \phi(x)^{k+1}$. In these calculations we use the laws of exponentiation. By induction, the result holds for all $n \geq 0$.
To show the result when $n = -1$, note $e_H = \phi(e_G) = \phi(x.x^{-1}) = \phi(x)\phi(x^{-1})$, whence $\phi(x^{-1}) = \phi(x)^{-1}$.
Now suppose $n = -m$ with $m > 1$. Then $\phi(x^n) = \phi(x^{-m}) = \phi((x^{-1})^m)$. We now apply our previous result regarding non-negative exponents to x^{-1}. Thus, $\phi((x^{-1})^m) = \phi(x^{-1})^m$. But, by the result just proved, $\phi(x^{-1}) = \phi(x)^{-1}$. So $\phi(x^n) = (\phi(x)^{-1})^m = \phi(x)^{-m} = \phi(x)^n$. In all these calculations, we use laws of exponents.

(b) If $\text{ord}(x) = n$ then $\phi(x^n) = \phi(x)^n = \phi(e_G) = e_H$, which implies $\text{ord}(\phi(x)) = k$ is finite and divides $n = \text{ord}(x)$. Indeed, write $n = kt + r$ with $0 \leq r < k$ (Euclidian division) and $y = \phi(x)$. Then $y^n = y^{tk+r} = y^{tk}y^r = (y^k)^t y^r = e_H^t y^r = e_H y^r = y^r = e_H$ which implies $r = 0$ by the definition of k.
If ϕ is an isomorphism consider $\phi^{-1} : H \to G$ which is a bijection. Then ϕ^{-1} is a homomorphism: if $h_1 = \phi(g_1)$ and $h_2 = \phi(g_2)$ are elements of H then $\phi^{-1}(h_1h_2) = \phi^{-1}(\phi(g_1)\phi(g_2)) = \phi^{-1}(\phi(g_1g_2)) = g_1g_2 = \phi^{-1}(h_1)\phi^{-1}(h_2)$. Thus ϕ^{-1} is an isomorphism (it is a bijection and a homomorphism). We conclude in a similar way as for ϕ above that $\text{ord}(x) = \text{ord}(\phi^{-1}(\phi(x))$ divides $\text{ord}(\phi(x))$ so they must be equal.

- Consider the homomorphism $\phi : \mathbb{Z} \to \mathbf{Z}_n$ defined by $k \mapsto [k]_n$, then $\text{ord}(1) = \infty$ in $\mathbb{Z}$, and $\text{ord}(\phi(1)) = \text{ord}[1]_n = n < \infty$.
- Consider, for a given integer $m \geq 1$, the homomorphism $\psi : \mathbb{Z} \to \mathbb{Z}$ defined by $\psi(k) = mk$. Then $\text{ord}(1) = \infty$ and $\text{ord}(\psi(1)) = \text{ord}(m) = \infty$.

7. Assume $ab = ba$ then $(ab)^n = ab \cdots ab = a \cdots ab \cdots b = a^n b^n$, where $\cdots$ means product n times, and the middle equality follows from the fact that a and b commute.